PEB WORK

TECHNICAL SPECIFICATION

FOR

PRE-ENGINEERED STEEL STRUCTURES

S. No	Name of Building/Shed	General Arrangement drawing Reference No.
1	Aircraft hanger	RITES/AP/MEA-2/ARCH/HANGER/01&02
2	Base Repair Workshop	RITES/AP/MEA-2/ARCH/REPAIR WORKS/01,02&03
3	Storage shed for Lubricating oil	RITES/AP/MEA-2/ARCH/STO.SHED-OIL/01
4	Storage Shed for Rubberized items/Spares	RITES/AP/MEA-2/ARCH/STO.RUBB/01
5	Refrigerated Shed	RITES/AP/MEA-2/ARCH/REF.SHED/01

The PEB Structure will consist of all components required to construct the primary and secondary framing (wind bents, wall & roof purlins) plus the wall and roof cladding. The contractor will complete the work on **Design & Build basis.Detailed design and drawing of structure including Foundation for Pre Engineered Building (as per general arrangement drawing) shall be provided by the contractor duly vetted by NIT / IIT or any approved agency of International repute (agency approved by Engineer – in charge) before the start of work for the approval of Engineer-in-charge.**

The contractor will also give a **10 year leak proof guarantee** (in the form of Guarantee Bond in respect of leak proof structure).for the pre-engineered building from the date of completion of work..The pre-engineered buildings shall be designed according to the following codes:

Design Codes

Seismic Load As per IS 1893 -2002

Wind Load IS -875

Design IS -800 -1984 (WORKING STRESS DESIGN METHOD) / Limit State method / IBC

Deflection limits to be taken from IS 800-2007/IBC (International building Codes)

Design Loads

Live Load- 75 kg /m^2

Live Load shall be as per IS-875.

Dead load

Self-Weight of the Structure including Purlins, Roof Sheeting, Girts Bracings, weight of turbo ventilators to be added as dead load.etc.

Solar panel load on roof structure- 25 Kg/m^2

Wind Load

Wind load shall be **300Km/hr.**

The design life of structure shall be taken as **50 years.**

Earthquake load

Seismic forces shall be as per IS: 1893, Zone V.

CBR value: - 5%

A. GENERAL FEATURES

Type:TCCS (Tapered Column Clear Span)

Width:as per General Arrangement Drawing

Length:as per General Arrangement Drawing

Eave height:as per General Arrangement Drawing

Width module: 1

Roof slope:min 1 in 10

Bay spacing:as per General Arrangement drawing

Front End Frame: Standard Post & Beam System

Back End Frame: Standard Post & Beam System

Type of bracing on roof &wall: Cold formed top hat section/Diagonal Angle bracing (as per design)

Wall panel:as per detailed product specification as under.

Girts: Sidewall girts and end wall girts are mounted on the exterior face of the main column.

Fasteners: Mechanically galvanized Hex-washer head self-drilling fasteners with integral EPDM seals.

Accessories: Flashing, trims, eave gutters & downspouts.

Canopy: As per General Arrangement drawing.

Self-propelled turbo-ventilation: 600 mm throat Dia with FRP base as approved

by Engineer-in-charge

Doors& windows : Aluminum Panel/ Aluminum Glazed

Paint Finish: All primary and secondary members shall be suitably primed&painted to **C5M Standards.**

Building Layout:as per General Arrangement Drawing

B. DETAILED PRODUCT SPECIFICATION

SHEETING

1. **The Roof sheeting for Refrigerated Shed:** Double Skin Roofing System i.e. linear sheet with sub-girts with Top Sheet of standing seam (0.60mm, 345Mpa) Color Coated Galvalume sheet (CCGV) and Bottom sheet of (0.47mm, 550 Mpa) Bare Galvalume sheet with 50mm thick x 24 kg/m3 density Glass Wool Insulation with one side aluminum foil & GI mesh and sub-girts.Color of Roofing Sheet shall be as approved by Engineer-charge.
2. **The Roof sheeting for Hanger/Workshop/Lubricating & Rubberized Sheds** : Standing seam roof sheeting (single skin) with 360 degree double-lock standing seam profile of color coated galvalume/zincalume sheet of 0.60mm base metal thickness &345Mpa yield strength as per approved make.In Workshop for natural light/skylight 2 mm thick Polycarbonate sheet of approved make shall be provided as per drawing and as directed by Engineer-in-charge. Color of Roofing Sheet shall be as approved by Engineer-charge
3. **Wall Sheeting for Building/Sheds**: 0.50 mm color coated galvalume/zincalume sheet (Trapezoidal profile) & 550 Mpa yield strength as per approved make. Color of Sheet shall be as approved by Engineer-charge. For natural light/skylight 2mm thickPolycarbonate sheet of approved make shall be provided as per drawing and as directed by Engineer-in-charge.

STRUCTURAL MEMBERS:

Frame: Supply of Primary (Built-up) sections, fabricated from hot rolled steel

plates. Structural Steel shall be Grade S355JOWP/S355JRWP conforming to EN: 10025-5:2004 PART 5 with minimum yield strength of 345 Mpa. Flanges are welded to the web by a continuous fillet weld deposited by an automatic submerged arc welding process on all four sides. The Built up frame shall be shot blast/sand blast &shall be suitably primed by zinc rich primer& painted as per **C5M** standards with two coats of brush applied paint at plant only.

Any damage to the paint during shifting of frame shall have to be made good at site suitably.

Supply of Galvanized secondary members, cold-formed from steel coils conforming to ASTM A 653 M Grade 65, with zinc coating to Z 275 designation (275 g/m^2) having a minimum yield strength of 450 Mpa.

ROOFING PANEL:

Supply of standing seam roofing system with 360 degree double-lock standing seam profile, effective cover width with two major corrugations, 50 mm high. The flat of the panel shall micro ribs to the major corrugations over the entire length of the panel. The feed material is manufactured from 0.60 mm Base Metal thickness (BMT), min. 345 MPa yield strength coated with hot dip metallic Aluminum /Zinc alloy coating, Zincalume AZ150 as 150 gms/ sq.mt total on both sides of Aluminum (55%) & Zinc (43.50 %) & Silicon (1.50%), conforms to ASTM792M or AS1397, roofing system .

The steel manufacturers test certificate for the chemical and mechanical properties of steel shall be submitted by the agency. The sheet shall have brand marking of the manufacturer on the back of the sheet at every 1 m c/c for confirming genuinity of the material. Specially designed roof clips shall be used to hold roof panels to the supporting structural member. The supporting structural member (galvanized purlins) shall be factory pre-punched as per design requirements. Panel side laps shall be field-seamed by roof runner seaming machine which is self-propelled and portable electrical lock-seaming machine. The machine field forms the final 180 degrees of a 360 degree double-lock standing seam. For Double Skin Roofing System i.e linear sheet

with sub-girts, Top sheet of standing seam(0.60mm,345Mpa) Color Coated Galvalume Sheet(CCGV) and Bottom Sheet of (0.47mm,550 Mpa) Bare Galvalume Sheet to be used. The insulation shall be vinyl membrane supported fiberglass blanket of thickness 50 mm with density 24 kg/m^3 (or as per design requirement). Accessories such as ridge capping, flashing; trims, gutter and down pipe shall be used for fixing roof system. Self drilling screws/fasteners shall be used for fixing roof system.

WALL PANELS:

Wall Panels are roll formed from nominal 0.50 mm base metal thickness of minimum yield strength of 550 MPa, coated with an aluminum /zinc alloy (i.e. Zincalume Steel), AZ150 (min 150 gm/m2 total on both side), conforms to Australian standard AS1397, pre-painted with Color bond steel quality paint coat as per AS/NZS 2728 Class 3 or any other equivalent panel. The paint finish thickness shall have a total coating thickness of nominal 35 µm, comprising of nominal 20 µm on exterior face and nominal 5 µm reverse coat on interior face over nominal 5 µm epoxy primer coat on both surfaces of approved colour shade by concern authority. The steel manufacturer's test certificate for the chemical and mechanical properties of steel shall be submitted by the agency. The sheet shall have brand marking of the manufacturer giving product details on the back of the sheet at every 1 meter c/c for confirming genuinity of the material.

PROFILE DIMENSIONS:

Wall panel with profile of 1000 mm effective cover width, min. 30 mm deep ribs at pitch of min. 200 mm centre to centre distance with two stiffeners between the ribs.

TRIMS&FLASHING: Wall flashing and trims (gable, corner, framed opening, accessories, etc.) are manufactured from same color, finish and thickness as wall panels.

Roof flashing and trims (parapet flashing, transition trims, expansion joint

trims and ridge caps) are manufactured from same color, finish and thickness as roof panels.

Roof water gutters will be made of minimum 2 mm thick GI (275 GSM) color coated material and downspouts are.of GI material and will be of same color as wall panels.

Water-resistant louvers shall be manufactured from high strength ZINCALUME steel/GI sheet.

PROTECTION ACCESSORIES:

Protection netting shall be provided.

OTHER ACCESSORIS:

Anchor bolts are manufactured from rods conforming to ASTM A 36M Grade 36 or equivalent with minimum yield strength of 240 MPa and an ultimate strength of 400 MPa.

Cold rolled angle/ top hat section for bracing members will be used in sidewalls of buildings supporting cranes conforming to ASTM A36 M or equivalent with minimum yield strength of 240 MPa.

Flange braces used to stabilize the inner flanges of main frame columns and raftersare GI turn buckle type with minimum yield strength of 240 MPa

Anti Sag Sections between purlins:-To be made out of minimum 2mm thick cold formed / hot rolled sections with end cleats having 02 no. of bolts.

Panel End lap: The panel lap shall be joined by means of a two-piece clamped connection consisting of a bottom reinforcing plate and a top panel strap as per approved design and to the satisfaction of Engineer-In-Charge. All other special accessories should also be factory fabricated including flashings, ridge cap, gutter, downpipe etc.

Sealant: Special grade of silicon sealant non-hardening, neutral cure type of approved make and grade shall be applied at all side laps and end laps (with flow ablemastic) as approved by engineer-in-charge.

Bead mastic is an extruded elastomeric butyl rubber based sealant supplied in rolls on silicon release paper conforming to Federal Specification TT-C-1796 A

Type II Class B (or equivalent).

Flow able mastic (caulking sealant) is a neutral cure silicone rubber sealant that is chemically inert and non corrosive. It is UV resistant and suitable for exterior applications against weathering and rainwater. When cured it is non-toxic and will accommodate high thermal and shrinkage changes in structural movement joints.

Foam closures match the panel profile. They are made of expandedpolyurethane or similar material.

Fasteners: The panel clip shall be fastened to structural members with Scrubolt fastener or equivalent material. The size of the fastener shall be as per the manufacturers' recommendation and as approved by engineer-in-charge.

Insulation: The insulation shall be vinyl membrane supported fiberglass blanket of thickness 50 mm with density min 24 kg/m^3 (or as per design requirement) and as approved by engineer-in-charge.

Specification for Skylight (Translucent sheeting):

2 mm thick polycarbonate sheet of approved make shall be provided.

Bird Protection Mesh: It shall be made with galvanized steel wire mesh of opening size25 x 25 mm made of 16 gauze minimum thickness of wire. The wire mesh to be fixed on frame of GI angles.

The work also includes design of following:

- Rigid frame structural for columns, rafters with end plates, Purlins, girts.
- Wind Operated Turbo Ventilators
- All necessary flashings, trims, eave gutters & downspout
- Roofing & cladding sheets
- Structural hardware & sheeting fasteners
- End lap sealant
- Bracing
- Anchor bolts and templates
- 2 mm thick FRP day light panel
- Framed opening

- Canopy system
- Interior partition wall, Doors and fixtures
- Wall vents, doors, windows and louvers
- Concrete embed plates of weld plate of any kind
- Design of Foundations
- Crane beam, brackets, crane rails and crane
- Electricity & Water required during work at site.
- Proof checking by a reputed institute IIT/NIT/or any approved agency of International repute

ERECTION &FIXING:

- The contractor will be required to submit design calculation in support of the proposed profile of the sheet and standard loading etc. The contractor shall also submit methodology for fixing and also a maintenance manual for routine maintenance.
- The installation shall be done in accordance to the standard practices as specified by the manufacturer and approved by Engineer-in-charge. All sheets and accessories must be stored and finally erected without any damage. No cutting of holes, welding of joints plates, fixing of shims will be allowed at site.
- Special flashing, ridge capping and trims shall be fixed as per manufacturer's recommendation. The shape and girths shall be as per design requirement
- Panel side lap shall be field-seamed by a self-propelled and portable electrical lock-seaming machine. The machine field forms the final 180 degrees of a 360 degree double-lock standing seam; all side lap sealant shall be factory applied.
- The contractor shall ensure that panel erector is familiarized with the erection procedure and all the supporting members are straight, level and true before starting panel erection. Panels shall be erected according to approved shop drawings by Engineer-in-Charge.

MEASUREMENT:

- **The Pre-engineered building shall be measured from outer to outer of roof overhang on both the sides. No separate payment shall be made for projected edge of canopy on any side.**
- No separate payment will be made for the laps of sheet and accessories. bolts, nuts, washers, adjustable bolts and supports for gutters and other fixtures. These are assumed to be included in the quoted rates.

Note: The contractor shall prepare the shop drawings based on the GAD drawings provided by the Engineer-in-Charge. These shall be submitted in five sets sufficiently in advance to the Engineer-in-charge for approval.

SCHEDULE OF PAYMENT

S No.	Payment Stage	% age of Quoted Amount
1	Approval of Detailed design and drawing of Structure including Foundation by Engineer– in charge.	10 %
2	Submission of Manufacturer Test Certificate and Receipt of entire material at Construction site.	55%
3	Erection of frame at site of work.	25%
4	Completion of erection & commissioning of Pre-engineered building including ancillary works as required.	10%

DESCRIPTION OFMATERIAL

1. **Primary Steel**

All primary steel members shall be solid web built-up section in manufactured

out of 345 Mpa steel plates. Welding of web to flange shall be SAW type on both sides.

2. **Cold formed section**

These shall be Z/C shaped cold formed section out of 345 Mpa Galvanized 275 GSM steel as per specification.

3. **Miscellaneous steel components**

Flange bracing, miscellaneous steel components such as clips etc shall be out of 240/250 Mpa steel conforming to IS 2062.

4. **Structural Hardware**

All primary steel connections shall be established using high tensile Gr. 8.8 fasteners. Anchor bolts and misc. steel connections shall be done by Galvanized Gr. 4.6 fasteners.

PRE- ENGINEERED STRUCTURES (APPROVED SUPPLIER):

- Cold Steel Corporation
- Tiger Steel
- TATA BLUESCOPE STEEL Ltd.
- KIRBY Engineering
- Hindustan Alcox Limited
- Speco Engineering Pvt.Ltd.

ROOF & WALL SHEETING (APPROVED SUPPLIERS):

- TATA BLUESCOPE STEEL Ltd.
- GE Plastics
- Danpalon
- Jindal

HANGER SPECIFICATIONS

PRE ENGINEERED BUILDING (SOLID WEB PORTALS - AS PER LAYOUT ATTACHED)	
SPAN OF BUILDING	57 Meters. c/c clear span
BUILDING SPACING X No. OF BAYS	7.5 Meters. c/c X 7
CLEAR HEIGHT OF BUILDING	18 Meters from FFL on grid line B & C
LENGTH OF BUILDING	52.5 Meters. c/c from grid 1 – 8
ROOF SLOPE	5.71 DEG. (1:10)
SPAN OF LEAN-TO	4 Meters c/c between grids A – B & C – D
LENGTH OF LEAN-TO	52.5 Meters c/c from grid 1 – 8
CLEAR HEIGHT OF LEAN-TO	3.5 Meters
ROOF OVERHANG FOR LEAN-TO	2.1 Meters c/c on grid line A & D from grid 1 - 8
STANDING SEAM FOR ROOF SHEETING **(Single Skin)**	**Standing seam of CCGV** (0.6 mm, 345 Mpa) has been considered only on roof area
SIDE SHEETING **(Single Skin)**	CCGV above 4.2 Meters from FFL (0.50 mm, 550 Mpa) on grid line B&C from grid 1 - 8 ,as per GAD drawing For skylight /natural light 2 mm thick polycarbonate sheet shall be provided above 4.2 Meters from F.F.L on grid line B&C from grid 1-8 as per GAD drawing.
GABLE SHEETING (Single Skin)	CCGV upto FFL (0.50 mm, 550 Mpa) has been considered on grid line 1 between grids A - D on rear end.
	CCGV upto FFL (0.50 mm, 550 Mpa) has been considered on grid line 8 between grids A–B'&C'D(Front) Sheeting also to be provided on the area over sliding door.

CRANE SPAN	37.02 Meters c/c
CRANE WHEELBASE	5 Meters
GANTRY BOTTOM HEIGHT	15.15 Meters
TURBO VENTILATORS	600mm Throat dia with FRP base
FRAMED OPENINGS	4 (W) X 4 (H) Meters – (01 no's)
DOOR OPENINGS	2 (W) X 3 (H) Meters – (02 no's)
CANOPY OVERHANG	2.5 Meters
CANOPY HEIGHT	4 Meters

MATERIAL SPECIFICATIONS	
MINIMUM THICKNESS OF H-BEAM FLANGE	6 mm.
MINIMUM THICKNESS OF H-BEAM WEB	4 mm.
MINIMUM THICKNESS OF COLD FORMED Z-SECTION	2 mm**(275 GSM, Galvanized)**

The Hanger structure to includefollowing members:-

1. Solid Web Portals With Lean-To
2. Gable End Frames
3. Roof Purlins With Accessories
4. Side Girts With Accessories
5. Gable Purlin With Accessories
6. Wind Bracing Crosses
7. 2 mm G.I. Rain Water Channel (with down take pipes)
8. Foundation Bolts
9. Nuts & Bolts
10. Standing Seam Of CCGV For Roof Sheets **(0.6mm) – Single Skin Sheeting**
11. CCGV Side Sheeting **(0.5mm, Single Skin Sheeting)**
12. CCGV Gable Sheeting **(0.5mm, Single Skin Sheeting)**
13. Gable Apron
14. Gutter Flashing & Anti Drip Flashings
15. Top Ridge

16. Crane Gantry (For 5 MT EOT)
17. Turbo Ventilators Of 600mm Throat Dia
18. Framed Opening Of Size 4 (W) X 4 (H) Meters
19. Door Opening Of Size 2 (W) X 3 (H) Meters
20. Canopy Of 2.5 Meters Overhang.

REPAIR WORKSHOP SPECIFICATIONS

PRE ENGINEERED BUILDING (SOLID WEB PORTALS - AS PER LAYOUT ATTACHED)	
SPAN OF BUILDING	42 Meters. c/c clear span
BUILDING SPACING X No. OF BAYS	6 Meters. c/c X 7
EAVES HEIGHT OF BUILDING	11.75 Meters from FFL on grid line B & I
LENGTH OF BUILDING	42 Meters. c/c from grid 2 – 9
ROOF SLOPE	5.71 DEG. (1:10)
STANDING SEAM FOR ROOF SHEETING **(Single skin)**	**Standing seam of CCGV** (0.6 mm, 345 Mpa) has been considered only on roof area For skylight/natural light 2 mm thick Polycarbonate sheet shall be provided as per GAD drawing.
SIDE SHEETING (Single skin)	CCGV above 8.5 Meters from FFL (0.50 mm, 550 Mpa) on grid line B & I from grid 2 – 9 For skylight/natural light 2 mm thick Polycarbonate sheet shall be provided as per GAD drawing.
GABLE SHEETING (Single skin)	CCGV above 6Meters from FFL (0.50 mm, 550 Mpa) on grid line 2 & 9 from grid B – I For skylight/natural light 2 mm thick Polycarbonate sheet shall be provided as per GAD drawing.
CRANE SPAN	30 Meters c/c
CRANE WHEELBASE	5 Meters
GANTRY TOP HEIGHT	6.5 Meters
TURBO VENTILATORS	600mm Throat dia with FRP base
CONTINUOUS CANOPY OVERHANG	6 Meters

CANOPY HEIGHT	6 Meters

MATERIAL SPECIFICATIONS	
MINIMUM THICKNESS OF H-BEAM FLANGE	6 mm.
MINIMUM THICKNESS OF H-BEAM WEB	4 mm.
MINIMUM THICKNESS OF COLD FORMED Z- SECTION	2 mm. **(275 GSM, Galvanized)**

The Repair Workshop structure to includefollowing members:-

1. Solid Web Portals
2. Gable End Frames
3. Roof Purlins With Accessories
4. Side Girts With Accessories
5. Gable Purlin With Accessories
6. Wind Bracing Crosses
7. 2 mm G.I. Rain Water Channel (with down take pipes)
8. Foundation Bolts
9. Nuts & Bolts
10. Standing Seam Of CCGV For Roof Sheets **(0.6mm) – Single Skin Sheeting**
11. CCGV Side Sheeting **(0.5mm, Single Skin Sheeting)**
12. CCGV Gable Sheeting **(0.5mm, Single Skin Sheeting)**
13. Gable Apron
14. Gutter Flashing & Anti Drip Flashings
15. Top Ridge
16. Crane Gantry
17. Turbo Ventilators Of 600mm Throat Dia
18. Continuous Canopy Of 6.0Meters Overhang

LUBRICANT SHED SPECIFICATIONS

PRE ENGINEERED BUILDING (SOLID WEB PORTALS - AS PER LAYOUT ATTACHED)	
SPAN OF BUILDING	18.75 Meters. c/c clear span
BUILDING SPACING X No. OF BAYS	7.29 Meters. c/c X 2 + 7.375 Meters c/c X 7
CLEAR HEIGHT OF BUILDING	3.27 Meters from FFL on grid line 1 & 5
LENGTH OF BUILDING	66.205 Meters. c/c from grid A – J
ROOF SLOPE	5.71 DEG. (1:10)
ROOF OVERHANG	Roof overhang of 1 Meters on grid line A & J from grid 1 - 5 and on grid line 1 from grid A - J
STANDING SEAM FOR ROOF SHEETING **(Single skin)**	**Standing seam of CCGV** (0.6 mm, 345 Mpa) has been considered only on roof area
SIDE & GABLE SHEETING (Single skin)	CCGV above 3 Meters from FFL (0.50 mm, 550 Mpa)
PARTITION SHEETING (Single skin)	CCGV above 3 Meters from FFL (0.50 mm, 550 Mpa) on grid line C, E & H between grids 1 – 5 on one side of column only.
TURBO VENTILATORS	600mm Throat dia with FRP base
FRAMED OPENINGS	3 (W) X 2.7 (H) Meters – (07 no's)
CONTINUOUS CANOPY OVERHANG	3 Meters
CANOPY HEIGHT	2.7 Meters

MATERIAL SPECIFICATIONS	
MINIMUM THICKNESS OF H-BEAM FLANGE	6 mm.
MINIMUM THICKNESS OF H-BEAM WEB	4 mm.
MINIMUM THICKNESS OF COLD FORMED Z- SECTION	2 mm. **(275 GSM, Galvanized)**

The Lubricant Shed Structure to includefollowing members:-

1. Solid Web Portals
2. Gable End Frames
3. Partition Columns
4. Roof Purlins With Accessories
5. Side Girts With Accessories
6. Gable Purlin With Accessories
7. Partition Purlin With Accessories
8. Wind Bracing Crosses
9. 2 mm G.I. Rain Water Channel (with down take pipes)
10. Foundation Bolts
11. Nuts & Bolts
12. Standing Seam Of CCGV For Roof Sheets **(0.6mm) – Single Skin Sheeting**
13. CCGV Side Sheeting **(0.5mm, Single Skin Sheeting)**
14. CCGV Gable Sheeting **(0.5mm, Single Skin Sheeting)**
15. CCGV Partition Sheeting **(0.5mm, Single Skin Sheeting)**
16. Gable Apron
17. Gutter Flashing & Anti Drip Flashings
18. Top Ridge
19. Turbo Ventilators Of 600mm Throat Dia
20. Framed Opening of Size 3(W) x 2.7(H) Meters
21. Continuous Canopy Of 3.0 Meters Overhang

RUBBERISED SHEDSPCIFICATIONS

PRE ENGINEERED BUILDING (SOLID WEB PORTALS - AS PER LAYOUT ATTACHED)	
SPAN OF BUILDING	18.75 Meters. c/c clear span
BUILDING SPACING X No. OF BAYS	7.29 Meters. c/c X 2 + 7.375 Mtrs c/c X 7
CLEAR HEIGHT OF BUILDING	3.27 Meters from FFL on grid line 1 & 5
LENGTH OF BUILDING	66.205 Meters. c/c from grid A – J
ROOF SLOPE	5.71 DEG. (1:10)
ROOF OVERHANG	Roof overhang of 1 Meters on grid line A & J from grid 1 – 5 and on grid line 1 from grid A – J
STANDING SEAM FOR ROOF SHEETING **(Single skin)**	**Standing seam of CCGV** (0.6 mm, 345 Mpa) has been considered only on roof area.
SIDE & GABLE SHEETING (Single skin)	CCGV above 3 Meters from FFL (0.50 mm, 550 Mpa)
PARTITION SHEETING (Single skin)	CCGV above 3 Meters from FFL (0.50 mm, 550 Mpa) on grid line C & F between grids 1 – 5 on one side of column only.
TURBO VENTILATORS	600mm Throat dia with FRP base
FRAMED OPENINGS	3 (W) X 2.7 (H) Meters – (05 no's)
CONTINUOUS CANOPY OVERHANG	3 Meters
CANOPY HEIGHT	2.7 Meters

MATERIAL SPECIFICATIONS	
MINIMUM THICKNESS OF H-BEAM FLANGE	6 mm.
MINIMUM THICKNESS OF H-BEAM WEB	4 mm.
MINIMUM THICKNESS OF COLD FORMED Z- SECTION	2 mm. **(275 GSM, Galvanized)**

Before Issue
Insertions (I).....
Corrections (CR).....
Cutting (CT).....
Omissions (OM).....
Over Writing (OW).....

NIL

366

After Receipt
Insertions (I).....
Corrections (CR).....
Cutting (CT).....
Omissions (OM).....
Over Writing (OW).....

The Rubberized Shed Structure to includefollowing members:-

1. Solid Web Portals
2. Gable End Frames
3. Partition Columns
4. Roof Purlins With Accessories
5. Side Girts With Accessories
6. Gable Purlin With Accessories
7. Partition Purlin With Accessories
8. Wind Bracing Crosses
9. 2 mm G.I. Rain Water Channel (with down take pipes)
10. Foundation Bolts
11. Nuts & Bolts
12. Standing Seam Of CCGV For Roof Sheets **(0.6mm) – Single Skin Sheeting**
13. CCGV Side Sheeting **(0.5mm, Single Skin Sheeting)**
14. CCGV Gable Sheeting **(0.5mm, Single Skin Sheeting)**
15. CCGV Partition Sheeting **(0.5mm, Single Skin Sheeting)**
16. Gable Apron
17. Gutter Flashing & Anti Drip Flashings
18. Top Ridge
19. Turbo Ventilators Of 600mm Throat Dia
20. Framed Opening of Size 3(W) x 2.7(H) Meters
21. Continuous Canopy Of 3.0 Meters Overhang

REFRIGERATED SHED SPECIFICATIONS

PRE ENGINEERED BUILDING (SOLID WEB PORTALS - AS PER LAYOUT ATTACHED)	
SPAN OF BUILDING	19.085 Meters. c/c clear span
BUILDING SPACING X No. OF BAYS	8.2 Meters. c/c X 4
CLEAR HEIGHT OF BUILDING	3.275 Meters from FFL on grid line 3 & 8
LENGTH OF BUILDING	32.8 Meters. c/c from grid A – E
ROOF SLOPE	5.71 DEG. (1:10)
ROOF SHEETING **(Double skin i.e. liner sheet with sub girts)**	Top sheet of Standing seam CCGV (0.6 mm, 345 Mpa) has been considered on roof area
	Bottom sheeting (screw down sheeting) of Bare GV 0.47 mm, 550 Mpa) has been considered for roof area
SIDE SHEETING (Single skin)	CCGV above 3 Meters from FFL (0.50 mm, 550 Mpa) on grid line 8 & 3 from grid A – E
GABLE SHEETING (Single skin)	CCGV above 3 Meters from FFL (0.50 mm, 550 Mpa) on grid line A & E from grid 3 – 8
TURBO VENTILATORS	600mm Throat dia with FRP base
CONTINUOUS CANOPY OVERHANG	3 Meters
CANOPY HEIGHT	2.25 Meters
FRAMED OPENING	1.2 (W) X 2.1 (H) Meters
GLASSWOOL INSULATION	Glass wool insulation of 50mm thick & 24 kg/m3 density with one side aluminum foil & GI mesh has been considered only for roof area.

MATERIAL SPECIFICATIONS	
MINIMUM THICKNESS OF H-BEAM FLANGE	6 mm.
MINIMUM THICKNESS OF H-BEAM WEB	4 mm.
MINIMUM THICKNESS OF COLD FORMED Z-SECTION	2 mm. **(275 GSM, Galvanized)**

The Refrigerated Shed Structure to includefollowing members:-

1. Solid Web Portals
2. Gable End Frames
3. Roof Purlins With Accessories
4. Side Girts With Accessories
5. Gable Purlin With Accessories
6. Wind Bracing Crosses
7. 2 mm G.I. Rain Water Channel (with down take pipes)
8. Foundation Bolts
9. Nuts & Bolts
10. Roof Sheets **(Double skin i.e linear sheet with sub girts) i.e standing seam CCGV 0.6 mm sheet on top & Bare GV 0.47mm on bottom**
11. CCGV Side Sheeting **(0.5mm, Single Skin Sheeting)**
12. CCGV Gable Sheeting **(0.5mm, Single Skin Sheeting)**
13. Gable Apron
14. Gutter Flashing & Anti Drip Flashings
15. Top Ridge
16. Turbo Ventilators Of 600mm Throat Dia
17. Framed Openings
18. Continuous Canopy Of 3.0 Meters Overhang
19. Glass Wool Insulation for Roof Area

RESS SPECIFICATIONS

Specifications for Rapid Erect Sun Shelter (RESS)

1. Scope of Work and Terms of Reference

The Scope of work shall include design, manufacture, transport to site from manufacturer's works, delivery, off load and move to erection site, on site assembly and erection, installation, testing, setting to work and commissioning of 1 No of Rapid Erect Sun Shelter (RESS).

Detailed design and drawing of structure including Foundation for RESS (as per general arrangement drawing) shall be provided by the contractor duly vetted by NIT/IIT or any approved agency of International repute (agency approved by Engineer – in charge) before the start of work for the approval of Engineer-in-charge.

2. Structural Performance Requirements – Design Codes

The proposed Facility is to be designed in accordance with the following design codes as applicable:

2.1 IS: 875 for various design loadings

2.2 IS: 1893 for seismic load

2.3 American Society of Civil Engineers (ASCE) 7, Minimum Design Loads for Buildings and Other Structures, 2005.

2.4 International Code Council, International Building Code 2006 Edition.

2.5 Aluminum alloy Design Manual: Specifications & Guidelines for Aluminum alloy Structures (The Aluminum alloy Association, 2005)

2.6 IS :800 for steel structures

2.7 IS : 456 for Reinforced Concrete

2.8 American Concrete Institute (ACI) ACI-318-95, Building Code Requirements for Reinforced Concrete.

2.9 American Welding Society (AWS), 2001 Structural Welding Code, Steel.

3. Environmental and Climatic Consideration

The following are the prevailing site conditions:

3.1 **Ambient temperature:**

	Minimum.	22.5^0C
	Max.	31.0^0 C

3.2 **Rainfall**

Annual Average,	160 mm
Monthly Average	309 mm (wettest month)

3.3 **Relative humidity**

Minimum	41 %
Maximum	100 %.

3.4 **Barometric pressure:** 1012 hPa.

3.5 **Wind:** General direction of wind is from the South East Direction

4. Design Parameters

4.1 **Dimensions**

Clear Span Width	53 m
Peak Height	17 m
Length 58 m	
Side Wall Cladding	Open up to 5.0 m from ground.

Gable end open at both ends

4.2 **Wind Speed Design Criteria**

Basic Wind Speed:	300 KMPH 3 Sec gust
Site Exposure:	"C" (open terrain)
Importance Factor:	Iw = 1

5. Standards for Design, Construction and Testing

5.1 **General**

The International System of measurement units (SI) shall apply throughout. The design, manufacture and testing of all elements of the Portable Covers together with all equipment and components of sub contract and/or external supply shall comply with the appropriate recommendations or requirements of the Indian Standards Institution.

5.2 **Where necessary, the following recognized International standards and codes are acceptable:**

a. Indian Standards Institute
b. Federal European de la Manutention (FEM)
c. British Standard Institute (BSI)
d. International Electro technical Commission (IEC)
e. Deutsches Institu fur Normunge.V(DIN)
f. American Standards

6. Design Loads & Criteria

The latest editions or equivalent of the following standards shall be used for the design of the RESS

6.1 **IS:875** : Design Loads (Other than earthquake) for buildings and structure

6.2 **IBC 2006** : International Building Code

6.3 **ASCE 7** : Minimum Design Loads for building & other structures

6.4 **IS800** : General Construction in Steel

6.5 **IS 1893** : Criteria for Earthquake Resistant Design of Structures

6.6 **Deflection:** The maximum deflection at any point on the structural framework shall not be more than 1/120 of the clear span width of the RESS when subjected to design load

7. Basis of Design

The basis of the design of the structural members shall be indicated in the technical offer including but not limited to the following:

7.1 Roof loading
7.2 Seismic zone
7.3 Zone factor
7.4 Importance factor
7.5 Response Reduction Factor
7.6 Panel Design
7.7 Load combinations in accordance with IS 800 and other relevant codes

8. Design Requirements

8.1 Structure to be designed to be of clear span, with no guy-ropes or cables extending or attaching perpendicular or at angles to the perimeter walls.

8.2 Structure to be designed so that there are no structural purlins, cables, or other obstructions to impede flow of water on the exterior skin panel.

8.3 All doors and other penetrations to have a separate frame, used for attachment of skin panel and designed to support the skin panel pre-stress and maximum working loads. Design these frames in such way that doors or other equipment inside them can be replaced without affecting the skin panel tension. **Direct attachment of skin panel to integral jambs is not acceptable.**

8.4 Structure to be designed & engineered so that no life-safety threat is created by the loss of any portion of either the interior or exterior skin panels. The structural framing shall not rely on the tensioning of the exterior skin panel for its stability or rigidity.

8.5 Skin panel shall be designed such that it meets the same load requirements as the structural frame, and the developed skin panel stresses must be applied to the structural frame. Allowable Stress Design (ASD) shall be used for skin panel stress verification with following minimum safety factor, using skin panel properties reduced for estimated loss of strength after 10 years.

8.6 The design and construction of the RESS should not allow perching of birds.

9. Design Life of the RESS

9.1 The structural members of the RESS shall have a **design life of 25 years** subject to "normal maintenance" of corrosion protection and mechanical & electrical components.

9.2 The light weight fabric cladding shall have a minimum life expectancy of **10 years** subject to "normal maintenance".

9.3 The Contractor shall provide a minimum **10 year** limited warranty for the skin panel.

9.4 The Contractor shall submit the "normal maintenance" that is forecasted for respective design life in terms of the materials and components and a schedule for the same.

10. Approved Drawings.

10.1 Three (3) copies of design, drawings including shop drawing along with work programs for erection, testing, commissioning and setting are to be provided to Engineer-in-charge for approval. The comments by Engineer-in-charge shall not relieve the Contractor of his responsibility for the correctness, adequacy of design and completeness of his work as per the Contract.

10.2 The Contractor shall provide to RITES, at delivery, three A1 size sets of prints of "as built" drawings, and two sets on a CD.

10.3 Manufacture's test certificate / test reports (from NABL Accredited Lab/ International repute testing labs) for Extruded box beams of Aluminum alloy, Skin Panel shall be provided by the contractor.

11. Construction

11.1 Construction of the RESS shall be to general good practice and to the applicable standards.

12. System Description.

12.1 Frames must consist of extruded box beams of aluminum alloy for their torsional stability characteristics. **I-Beam and Steel / Aluminum alloy truss frames are not acceptable.**

12.2 The frame supported, non-distortional special skin panel structure shall be a pre-engineered system of aluminum alloy framing components and skin panels

12.3 All materials shall be new. Used material is not acceptable.

12.4 Interior dimensions are to be maximized; therefore a single beam system of maximum 18" to 26"(±0.8") depth (preferably)should be used.

12.5 Primary framing shall be segmented, structural components designed to accept and secure the exterior skin panel. Framing shall be pre-engineered and shipped to the site as finished parts, to be assembled on site with appropriate hardware. Any kind of cutting & welding must be avoided. However, minor finishing work based on site requirements, involving drilling etc. can be permitted

12.6 Exterior skin panel will be panelized and run continuous over structure with integral kedar system edge for post-installation tensioning in both horizontal and vertical directions. Draped or loose laid skin panel

installation is not acceptable. A manufactured rope edge is not acceptable.

12.7 The groove (or track) to accept the keder system edge shall be part of the extruded profile.

12.8 The **colour** of the exterior skin panel will be as approved by Engineer-in-charge.

12.9 Exterior skin panel shall, upon completion, provide continuous, water and weather-tight enclosure.

12.10 Upon installation and tensioning, the exterior skin panel will be smooth and wrinkle-free and shall remain so under anticipated thermal, wind, snow and live-load conditions.

12.11 Skin panel shall be removable for repair or replacement with the least practicable disturbance to adjacent panels and with minimal need to dismantle structural framing.

12.12 Steel Cables wherever used must be of minimum diameter **16mm**(± 1.6mm) must be used.

\

13. Design Layout

13.1 Envisaged design layout of RESS is as per General Arrangement Drawing attached with the tender.

14. Material of Construction

14.1 **Structural Frame**

Structural frame shall be made of extruded box beams in Aircraft grade Aluminum Alloy as per ASTM B221, alloy 6061 T6 temper or alloy 6082 T6 temper.

14.2 **Exterior Skin**

a. The entire exterior skin panel shall be PVC minimum 650 g/m2. Skin panels should be of non-distortional, rip resistant material.

b. Skin panel used shall be **opaque.**

c. Exterior Skin panel shall be continuous from the base of the structure, over the structure.

d. All structural skin panel welds shall be completed using RF welding completed in manufacturing shop. Repair sealing and finishing work in the field to be kept to the minimum. If required, appropriate hand-welding equipment may be used.

All welds, shop or field, shall develop at least 90% (ninety per cent) of the base skin panel strength.

14.3 Skin panels shall be UL-or NFPA-701 rated flame-resistant materials. Manufacture's test certificate / **test reports (from NABL Accredited Lab / International repute testing labs) for Skin Panel shall be provided by the contractor.**

14.4 **Secondary Members to be used for connection**

a. Structural/Plates/Sheets
Wherever required contractor shall select suitable material **(confirming to BIS Standards)** with suitable protection against corrosion.

14.5 **Steel used for structural connections shall be as per standard below in line with design requirements -**

a.	Plates	IS 2062
b.	Steel Tubes - CHS	IS 1161
c.	Steel Tubes – RHS/SHS	IS 4923
d.	Bolts – Gr 8.8(Min)	IS 4000
e.	Anchor Bolts, Gr 5.6(Min)	IS 4000

14.6 **Cables**

a. Min diameter of the steel cable shall be 16mm.

b. ASTM A603 wire rope or A586 structural strand cables; use only one type with one modulus of elasticity throughout the structure.

c. Minimum breaking strength shall be at least double of the design load.

d. Complete assembly with fittings shall be tested & certified for the breaking load.

14.7 **Corrosion Protection of Steel**

a. All steel members used as connectors and/or fastening shall be hot dip galvanized to

I. Steel as per ASTM A123 (or ASTM A123M)

II. Hardware as per ASTM 153 (or ASTM A153M)

15. Foundations

15.1 All Civil work for foundation preparation to be carried out by Contractor as per approved drawing& paid in separate item as per BOQ.

15.2 Structure must be of ready to erect type.

16. Erection

16.1 Contractor must get RESS erected from approved vendor with scope of services including erection of the structure as well as of various system components, special tools and tackles as required.

17. Special Warranties

17.1 The Contractor shall provide a minimum 10 year limited warranty for the skin panel.

18. Electrification

18.1 Uniform lux intensity at 150 lux at ground level. Contractor to provide electrical lighting plan for achieving this level of lux intensity.

19. List of Approved vendors.

The work of **Rapid Erect Sun Shelter** is to be carried out by approved Vendors only.

1) Double Dee Technologies Pvt. Ltd.
(Aviation Structures Division)
Maneckji Wadia Building
127 Mahatma Gandhi Road,
Fort, Mumbai 400 023

2) Altair Industrial Technologies Pvt. Ltd.
8, Kumar Elite, Lane 4A Koregaon Park
Pune 411001

3) ABRO Technologies Ltd.
12, Scindia House, Connaught Circus
New Delhi 110001

20. MEASUREMENT:

RESS shall be measured from the outer face of Structural column to outer face of structural column on both the sides measured in Plan.

No separate payment will be made for the laps of sheet and accessories, bolts, nuts, washers, adjustable bolts and supports for gutters and other fixtures. These are assumed to be included in the quoted rates.

21. Payment Terms for RESS (Stage Payment for this item)

Stage	Stage of Completion	Amount to be released
1	**Approval of Detailed design and drawing of Structure i/c Foundation by Engineer– in charge.**	**10 % of the quoted rate**
2	**Preparatory work of Foundations & Footings for erection of Structure**	**10 % of the quoted rate.**
3	**Receipt of inspected material at site.**	**40 % of the quoted rate**
4	**Erection of Portal frames of RESS at site**	**20 % of the quoted rate.**
5	**Completion of erection & commissioning of RESS at site**	**20 % of the quoted rate**

The Cost of foundation /RCC shall be measured & paid separately under BOQ Item.